NEET UG 2027: Master Biology Chapter Notes

Genetics & Molecular Basis of Inheritance

Prepared, Compiled & Authored by Satya Prakash · High-Yield NCERT Master Notes

1 Principles of Inheritance & Variation (Mendelian Genetics)

Mendel's Experimental Architecture & Contrasting Traits

Gregor Johann Mendel conducted hybridization experiments on garden peas (*Pisum sativum*) for 7 years (1856–1863). He investigated **7 pairs of contrasting traits** located across 4 chromosomes (Chromosomes 1, 4, 5, 7):

Character	Dominant Trait	Recessive Trait	Chromosome
Stem Height	Tall (T)	Dwarf (t)	4
Flower Color	Violet / Purple (W)	White (w)	1
Flower Position	Axial (A)	Terminal (a)	4
Pod Shape	Inflated / Full (I)	Constricted (i)	4
Pod Color	Green (G)	Yellow (g)	5
Seed Shape	Round (R)	Wrinkled (r)	7
Seed Color	Yellow (Y)	Green (y)	1

Mendelian Ratios & Mathematical Formulations

For a cross involving heterozygous parents ($Aa \times Aa$ for monohybrid, $AaBb \times AaBb$ for dihybrid):

Monohybrid Phenotypic Ratio: $3:1$ (Genotypic: $1:2:1 = 1\,AA : 2\,Aa : 1\,aa$)

Dihybrid Phenotypic Ratio: $9:3:3:1$ (Genotypic: $1:2:1:2:4:2:1:2:1$)

Trihybrid Phenotypic Ratio: $(3+1)^3 = 27:9:9:9:3:3:3:1$

Number of Phenotypes: 2^n (n = number of heterozygous gene pairs)

Number of Genotypes: 3^n

Number of Gamete Types: 2^n

Number of Zygotes / Offspring: 4^n

Test Cross Ratio: Heterozygote crossed with homozygous recessive:

- Monohybrid Test Cross ($Aa \times aa$): Ratio $= 1:1$.
- Dihybrid Test Cross ($AaBb \times aabb$): Ratio $= 1:1:1:1$.

Deviations from Pure Mendelian Inheritance

1. **Incomplete Dominance:** Neither allele is dominant; the heterozygote displays an intermediate phenotype.
   - *Examples:* Flower color in *Mirabilis jalapa* (4 o'clock plant) and *Antirrhinum majus* (Snapdragon / Dog flower).
   - Crossing Red (RR) × White (rr) gives Pink (Rr) in F_1.
   - F_2 Phenotypic ratio = 1 Red : 2 Pink : 1 White ($1:2:1$).
   - **Crucial Rule: Phenotypic ratio equals Genotypic ratio** ($1:2:1$).
2. **Codominance:** Both alleles express themselves fully and simultaneously in the heterozygote without blending.
   - *Examples:* ABO Blood grouping in humans, Sickle cell trait ($Hb^A Hb^S$), MN blood group.
3. **Multiple Allelism:** More than two alleles govern a single gene locus in a population (can only be studied in population studies, not in an individual).
   - *ABO Blood Group Gene (I):* Located on Chromosome 9; 3 alleles (I^A, I^B, i).

- Number of possible genotypes for n multiple alleles:

$$\text{Total Genotypes} = \frac{n(n+1)}{2} = \frac{3(3+1)}{2} = 6 \text{ genotypes} \implies 4 \text{ phenotypes (A, B, AB, O)} \quad (1)$$

4. **Pleiotropy:** A single gene influences multiple unrelated phenotypic traits.
   - *Phenylketonuria (PKU):* Mutation in gene coding for enzyme *phenylalanine hydroxylase* causes mental retardation, reduction in hair, and skin pigmentation.
   - *Starch Synthesis in Pea Seeds:* Gene B controls starch grain size (incomplete dominance: BB large, Bb intermediate, bb small) and seed shape (complete dominance: BB, Bb round, bb wrinkled).
5. **Polygenic Inheritance:** Quantitative inheritance where multiple additive genes control a continuous trait, influenced by the environment (e.g., Human skin color, human height, kernel color in wheat).
   - Governed by 3 genes (A, B, C): $AABBCC$ (darkest) to $aabbcc$ (lightest).
   - F_2 phenotypic distribution follows a bell-shaped Gaussian curve: $1:6:15:20:15:6:1$.

Chromosomal Theory, Linkage & Genetic Mapping

1. Chromosomal Theory of Inheritance (Sutton & Boveri, 1902): Behavior of chromosomes parallels the behavior of genes during meiosis. Chromosomes segregate during anaphase I and assort independently during anaphase I/metaphase I.

2. Morgan's Linkage Experiments (Thomas Hunt Morgan, 1910): Used the fruit fly *Drosophila melanogaster* because:

- Can be grown on simple synthetic medium in the laboratory.
- Short life cycle (about 2 weeks / 14 days).
- A single mating produces a large number of progeny flies.
- Clear morphological dimorphism (female is distinctly larger than male).
- Low chromosome number ($2n = 8$) easily observable under low power microscope.

Morgan's Crosses on X-Chromosome:

- **Cross A (Tightly Linked):** Yellow body (y), white eye (w) crossed with wild brown body (y^+), red eye (w^+). Recombination frequency = **1.3%** (Parental types = **98.7%**).
- **Cross B (Loosely Linked):** White eye (w), miniature wing (m) crossed with wild red eye (w^+), normal wing (m^+). Recombination frequency = **37.2%** (Parental types = **62.8%**).

3. Chromosome Mapping (Alfred Sturtevant): Recombination frequency is directly proportional to the physical distance between genes on a chromosome:

$$\text{Distance between genes (in centimorgans / map units)} = \frac{\text{Total Recombinants}}{\text{Total Progeny}} \times 100\% \quad (2)$$

$$1 \text{ Map Unit (cM)} = 1\% \text{ Recombination frequency} \quad (3)$$

Boundary Limit: Maximum observable recombination frequency for two linked genes cannot exceed **50%** (at 50%, genes behave as if assorting independently).

Sex Determination Mechanisms in Animals & Humans

- **XX–XY Type (Male Heterogamety):** Humans and *Drosophila*. Male produces two types of sperms: 50% with X, 50% with Y. Female is homogametic (XX).
- **XX–XO Type (Male Heterogamety):** Grasshopper and certain insects. Male has only one X-chromosome $(2A + XO)$; female has two $(2A + XX)$.
- **ZZ–ZW Type (Female Heterogamety):** Birds, fowls, butterflies. Female produces two types of eggs: 50% with Z, 50% with W $(2A + ZW)$. Male is homogametic $(2A + ZZ)$.
- **Haplodiploidy (Honeybee):** Female (Queen/Worker) develops from fertilized diploid egg $(2n = 32)$ via sexual reproduction. Male (Drone) develops from unfertilized haploid egg $(n = 16)$ by **parthenogenesis**. Drones have no father and cannot have sons, but have a grandfather and can have grandsons!

NEET Trap: Genetic Disorders & Pedigree Distinctions

1. **Mendelian Disorders (Single-Gene Mutations):**

- **Autosomal Dominant:** *Myotonic dystrophy*, Huntington's chorea. Trait does not skip generations; affected child always has at least one affected parent.
- **Autosomal Recessive:** *Sickle-cell Anemia, Phenylketonuria, Thalassemia, Cystic Fibrosis.* Trait frequently skips generations; parents of affected individuals are usually unaffected heterozygous carriers (Aa). Both sexes equally affected.
  - **Sickle-cell Anemia:** Point mutation in β-globin gene on chromosome 11. GAG codon mutates to GUG $\implies$ substitution of Glutamic acid (Glu) by **Valine (Val) at position 6** of the β-chain. Under low O_2 tension, Hb^S polymerizes into elongated sickle shapes.
  - **Thalassemia:** Quantitative disorder (reduced rate of globin chain synthesis), whereas Sickle-cell anemia is a qualitative disorder (defective globin produced). α-Thalassemia is controlled by two closely linked genes $HBA1$ and $HBA2$ on chromosome 16. β-Thalassemia is controlled by a single gene HBB on chromosome 11.
- **Sex-Linked (X-Linked) Recessive:** *Hemophilia, Color Blindness.*
  - Show **Criss-cross inheritance** (passed from carrier mother to son, or father to daughter who becomes a carrier).
  - Far more prevalent in males (X^cY) than females (X^cX^c, requiring carrier mother and affected father).
  - **Hemophilia A:** Deficiency of blood clotting Factor VIII; **Hemophilia B:** Factor IX (Christmas disease).

2. **Chromosomal Disorders (Aneuploidy & Polyploidy):**

- **Down's Syndrome (Trisomy 21):** First described by Langdon Down (1866). Karyotype $= 47, XX+21$ or $47, XY + 21$. Short stature, small round head, furrowed tongue, partially open mouth, broad palm with characteristic palm crease, physical and mental retardation.
- **Klinefelter's Syndrome (Trisomy of Sex Chromosome):** Karyotype $= 47, XXY$. Overall masculine development with feminine features (e.g., development of breasts / **Gynaecomastia**), sterile male.
- **Turner's Syndrome (Monosomy of Sex Chromosome):** Karyotype $= 45, XO$. Sterile female, rudimentary ovaries, lack of secondary sexual characters, short stature, webbed neck.

2 Molecular Basis of Inheritance (DNA, Replication, Transcription & Translation)

Watson-Crick B-DNA Structural Dimensions

Proposed by James Watson and Francis Crick in 1953, based on X-ray diffraction data of Rosalind Franklin and Maurice Wilkins.

Helix Type: Right-handed double helix with antiparallel strands ($5' \to 3'$ and $3' \to 5'$)

Helix Pitch (P) **:** $3.4\,\text{nm} = 34\,\text{Å}$ (one complete turn of helix)

Base Pairs per Turn: $10\,\text{bp}$

Distance between bp: $0.34\,\text{nm} = 3.4\,\text{Å} = 0.34 \times 10^{-9}\,\text{m}$

Helix Diameter: $2.0\,\text{nm} = 20\,\text{Å}$

Hydrogen Bonds: A = T (2 H-bonds), G ≡ C (3 H-bonds)

Human DNA Length: $6.6 \times 10^9\,\text{bp} \times 0.34 \times 10^{-9}\,\text{m} \approx \mathbf{2.2}\,\text{m}$

Packaging of DNA: Nucleosome Architecture

Because a 2.2 m long negatively charged DNA must fit inside a microscopic nucleus ($\sim 10^{-6}$ m), it is tightly packaged around basic proteins called **histones**:

- **Histones:** Rich in basic amino acids **Lysine** and **Arginine** (both carry positive charges in their side chains).
- **Histone Octamer:** Core consists of two molecules each of H_2A, H_2B, H_3, and H_4.

- **Nucleosome Core:** Negatively charged DNA is wrapped around the octamer:

$$\text{Length of DNA wrapped around single nucleosome} \approx \mathbf{200}\,\text{bp} \tag{4}$$

- **Linker Histone (H_1):** Seals the DNA at the entry and exit points of the nucleosome core.
- **Chromatin States:**
  - **Euchromatin:** Loosely packed, stains light, transcriptionally active.
  - **Heterochromatin:** Densely packed, stains dark, transcriptionally inactive.

Historic Experiments Proving DNA is the Genetic Material

1. **Griffith's Transforming Principle (Frederick Griffith, 1928):** Used *Streptococcus pneumoniae* (Pneumococcus).
   - Smooth (S) strain has mucous (polysaccharide) coat $\implies$ Virulent (mice die of pneumonia).
   - Rough (R) strain lacks capsule $\implies$ Non-virulent (mice survive).
   - Heat-killed S strain $\rightarrow$ Mice survive.
   - Heat-killed S strain + Living R strain $\rightarrow$ **Mice die**, and living S strain bacteria were recovered from blood!
   - *Conclusion:* R strain was transformed by a heat-stable transforming principle from heat-killed S.
2. **Biochemical Characterization (Avery, MacLeod & McCarty, 1944):** Purified biochemicals (proteins, RNA, DNA) from heat-killed S cells.
   - Proteases and RNases did **not** inhibit transformation.
   - Digestion with **Deoxyribonuclease (DNase) completely inhibited transformation**.
   - *Conclusion:* DNA is the substance causing transformation!
3. **Unequivocal Proof: Hershey-Chase Experiment (1952):** Used Bacteriophage T2 and *Escherichia coli*.
   - Phages grown on radioactive sulfur (^{35}S) labeled protein coat (cysteine and methionine contain S, DNA has no S).
   - Phages grown on radioactive phosphorus (^{32}P) labeled DNA core (DNA has P, proteins have no P).
   - Steps: (i) Infection $\rightarrow$ (ii) Blending (agitation in kitchen blender detached coats) $\rightarrow$ (iii) Centrifugation.
   - Bacteria infected with ^{32}P phages were radioactive (pellet had radioactivity). Supernatant with ^{35}S had viral ghosts.
   - *Conclusion:* DNA is the genetic material passed from virus to bacteria!

DNA Replication Machinery & Enzymology

Replication is **semiconservative** and **bidirectional**, originating at specific sites called *Ori* (Origin of Replication).

Meselson & Stahl: Grew *E. coli* in $^{15}NH_4Cl \rightarrow$ all heavy ($^{15}N - ^{15}N$).

In $^{14}NH_4Cl$: Gen 1 (20 min): 100% Hybrid ($^{15}N - ^{14}N$).

Gen 2 (40 min): 50% Hybrid ($^{15}N - ^{14}N$) + 50% Light ($^{14}N - ^{14}N$).

Taylor: Tritiated thymidine in *Vicia faba* root cells.

Core Enzymes and Functional Roles:

Enzyme	Specific Biological Action
Helicase	Unwinds the parental DNA double helix by breaking hydrogen bonds, forming replication fork.
SSB Proteins	Single-Strand Binding proteins stabilize single strands and prevent premature re-annealing.
Topoisomerase / Gyrase	Relieves positive supercoiling strain ahead of the progressing replication fork by cutting and resealing.
RNA Primase	Synthesizes short complementary RNA primers ($5 \rightarrow 10$ nucleotides) providing free $3'$-OH for DNA polymerase.
DNA Polymerase III	Main replicating enzyme in prokaryotes; synthesizes in strictly $5' \rightarrow 3'$ **direction**; high catalytic speed (≈ 2000 bp/sec) with proofreading ($3' \rightarrow 5'$ exonuclease activity).
DNA Polymerase I	Removes RNA primers (via $5' \rightarrow 3'$ exonuclease activity) and fills gaps with deoxynucleotides (Kornberg enzyme).
DNA Ligase	Catalyzes formation of phosphodiester bonds between discontinuous Okazaki fragments on the lagging strand.

Transcription: Prokaryotes vs Eukaryotes

A transcription unit comprises: (1) **Promoter** (located towards $5'$ end of coding strand / upstream), (2) **Structural gene**, (3) **Terminator** (located towards $3'$ end of coding strand / downstream).

$$\text{Template Strand: } 3' \longrightarrow 5' \qquad \text{Coding Strand (Non-template): } 5' \longrightarrow 3' \tag{5}$$

Convention: All reference points (promoter, terminator) are defined with respect to the **coding strand**.

1. Transcription in Prokaryotes: Single RNA polymerase conducts all phases:

- Initiation: RNA Polymerase binds to promoter with transient assistance of **Sigma (σ) factor**.
- Elongation: RNA Polymerase synthesizes RNA in $5' \rightarrow 3'$ direction using ribonucleoside triphosphates.
- Termination: Termination factor **Rho (ρ)** associates with polymerase to terminate transcription.
- Translation can begin **coupled with transcription** even before mRNA is fully synthesized (no nuclear membrane separation).

2. Transcription in Eukaryotes (Three Polymerases & Post-Transcriptional Processing):

- **RNA Polymerase I:** Transcribes rRNAs ($28S, 18S$, and $5.8S$).
- **RNA Polymerase II:** Transcribes precursor of mRNA, called **heterogeneous nuclear RNA (hnRNA)**.
- **RNA Polymerase III:** Transcribes **tRNA**, $5S$ **rRNA**, and **snRNAs** (small nuclear RNAs).

Post-Transcriptional Modifications of hnRNA $\rightarrow$ mature mRNA:

1. **Capping:** An unusual nucleotide, **methyl guanosine triphosphate (m^7Gppp)**, is added to the $5'$-end.
2. **Tailing (Polyadenylation):** Adenylate residues ($200 - 300$) are added at $3'$-end in a template-independent manner.
3. **Splicing:** Non-coding intervening sequences (**Introns**) are excised, and coding sequences (**Exons**) are joined by spliceosomes (complex of snRNPs).

The Genetic Code: Salient Features & Wobble Phenomenon

Deciphered by George Gamow, Har Gobind Khorana, Marshall Nirenberg, and Severo Ochoa (polynucleotide phosphorylase).

Triplet Nature: 64 codons (4^3); 61 code for 20 amino acids; 3 stop codons.
Stop Codons: **UAA** (Ochre), **UAG** (Amber), **UGA** (Opal).
Dual Function: **AUG** codes for Met AND acts as initiator codon.
Degenerate: Amino acids coded by > 1 codon (Leu, Ser, Arg have 6).
Exceptions: AUG (Met) and UGG (Trp) have single codons alone.
Unambiguous: One specific codon codes for strictly one amino acid.
Universal: From bacteria to humans, UUU codes for Phe.

Wobble Hypothesis (Crick, 1966): The third base ($3'$ position of codon) does not pair as strictly with the first base ($5'$ position of anticodon) on tRNA. Non-Watson-Crick base pairing allows a single tRNA species

to recognize multiple degenerate codons, reducing the minimum required tRNAs to 31.

Translation: Mechanism of Polypeptide Synthesis

1. **Charging of tRNA (Aminoacylation):**

$$\text{Amino Acid} + \text{ATP} \xrightarrow{\text{Aminoacyl-tRNA synthetase}} \text{Aminoacyl-AMP} + \text{PP}_\text{i} \tag{6}$$

$$\text{Aminoacyl-AMP} + \text{tRNA} \longrightarrow \text{Aminoacyl-tRNA} + \text{AMP} \tag{7}$$

2. **Initiation:** Small ribosomal subunit ($30S$ in prokaryotes, $40S$ in eukaryotes) binds to mRNA at $5'$-end (Shine-Dalgarno sequence in prokaryotes). Initiator tRNA binds to P-site (Peptidyl site).
3. **Elongation:** Incoming charged tRNA binds to the A-site (Aminoacyl site). Peptide bond formation is catalyzed by the ribozyme:

$$\text{Peptidyl transferase} = \textbf{23S}\ \text{rRNA (in prokaryotes)} \quad \text{or} \quad \textbf{28S}\ \text{rRNA (in eukaryotes)} \tag{8}$$

   Ribosome advances codon by codon along mRNA in $5' \to 3'$ direction (**Translocation**, requires EF-G and GTP).
4. **Termination:** Stop codon arrives at A-site; release factors (RF1, RF2, RF3) bind to stop codon, hydrolyzing polypeptide from tRNA.

Gene Regulation: The Operon Model of Jacob & Monod

The **Lac Operon** of *E. coli* is an example of an **inducible, negative regulation system**:

- **Gene i (Inhibitor / Regulatory):** Constitutively expresses the active Lac repressor protein.
- **Promoter (p):** Binding site for RNA polymerase.
- **Operator (o):** Binding site for the Lac repressor protein.
- **Structural Genes:**
  - **Gene z:** Codes for **β-galactosidase** (hydrolyzes lactose into glucose and galactose).
  - **Gene y:** Codes for **Permease** (increases cellular membrane permeability to β-galactosides).
  - **Gene a:** Codes for **Transacetylase** (transfers acetyl group from acetyl-CoA to β-galactosides).
- **Regulation States:**
  - *In Absence of Inducer (Lactose):* Repressor binds tightly to operator $\implies$ RNA polymerase blocked $\implies$ Operon switched OFF.
  - *In Presence of Inducer (Lactose / Allolactose):* Inducer binds to repressor, inducing conformational change $\implies$ repressor inactivated and detaches $\implies$ RNA polymerase transcribes z, y, a $\implies$ Operon switched ON.

Human Genome Project (HGP) & DNA Fingerprinting High-Yield Facts

1. Human Genome Project Core Metrics:

- Total base pairs: 3.1647×10^9 bp; average gene contains **3000 bases**.
- Largest known human gene: **Dystrophin** on X-chromosome (2.4×10^6 bases).
- Total estimated genes: $\approx 30,000$ (drastically lower than previous estimates of 140,000).
- Less than **2% of the genome** codes for proteins.
- Chromosome 1 has the most genes (**2968**); the Y chromosome has the fewest (**231**).
- Scientists have identified about **1.4 million locations** where single-base DNA differences (**SNPs** – single nucleotide polymorphisms) occur.

2. DNA Fingerprinting (Alec Jeffreys, 1984):

- Explores **DNA polymorphism** in non-coding repetitive DNA sequence (**Satellite DNA**).
- **VNTRs (Variable Number of Tandem Repeats):** Belong to a class of satellite DNA called *minisatellites*. The copy number varies from chromosome to chromosome in an individual (size varies from 0.1 to 20 kb).
- **Protocol Steps:** Isolation of DNA $\to$ Digestion by restriction endonucleases $\to$ Separation by agarose gel electrophoresis $\to$ Southern blotting (transfer to nylon membrane) $\to$ Hybridization using labeled VNTR

probes → Detection by autoradiography.